

GEOG 172

What is Spatial Data?

This class introduces spatial data by having students first create spatial observations and then interpret mapped geographic patterns. The lecture is delivered by having students participate in two activities.

Lecture 1

Recall from last week....

We each added places we know and places we would like to know to a shared map and created a spatial dataset together.

We discussed how the pattern on that map depended on who happened to be in this room, where we have lived or traveled, or even what you have read or seen online/social media.

Crash Course on GeoAtom

The GeoAtom

the atomic form of geographic information (Goodchild 1999)

The GeoAtom

$$\langle \mathbf{s}, Z, z(\mathbf{s}) \rangle$$

where

$\mathbf{s}$ = a point in space--time,

Z = a property being observed,

$z(\mathbf{s})$ = the observed value of that property at $\mathbf{s}$.

For example, a geo-atom might indicate that at 120°W, 34°N, at 0 m above mean sea level, and at local noon on 11 July 2005 (a four-dimensional definition of $\mathbf{s}$), the Celsius temperature (the property Z) was 20° (the value $z(\mathbf{s})$).

The GeoAtom

A geographic observation brings together three basic pieces of information:

- where something is,
- what is there or what value we observe, and
- when that observation applies

We are going to build these pieces up gradually over the first few weeks of class!

Pin It to Win It



GeoGuesser is a geographic guessing game in which players are placed in an unknown location and attempt to determine where they are using only the clues visible in the surrounding landscape.

GeoGuesser Rules

Students work in groups of three and receive a worksheet and four stickers. Each group need to quickly look at the picture displayed on the board and guess where this picture is taken.

- You need to use visible geographic clues visible to guess where this picture might be taken.
- You may also use Google Maps / Street View only to reason about regions, not to directly search for the answer.

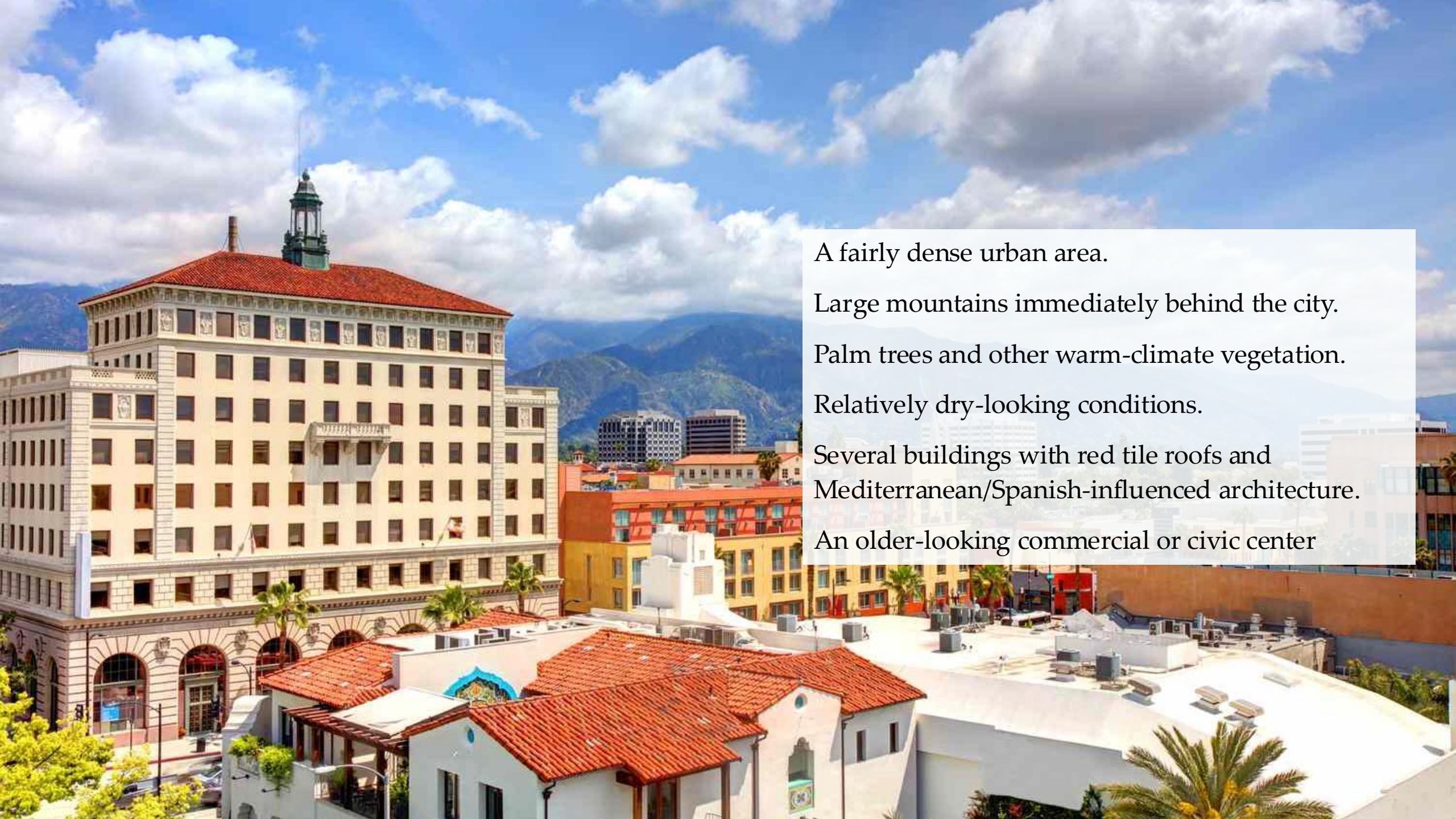
Each group must commit to one location per image. Once you've decided on the location, place a sticker on the grid. You will use a different color sticker for different pictures. You will also label the coordinates below your sticker.

Four minutes for each round! After time is up for each group need to report back the coordinate they guess. **The group that gets the closet answer will get a prize!**

At the end of this lecture, each group needs to submit to me a completed map with four stickers and the coordinates labeled below it, and **write down the name of the people they work with in the group (for attendance and next week's activity:)**

Let's work through an example!





A fairly dense urban area.

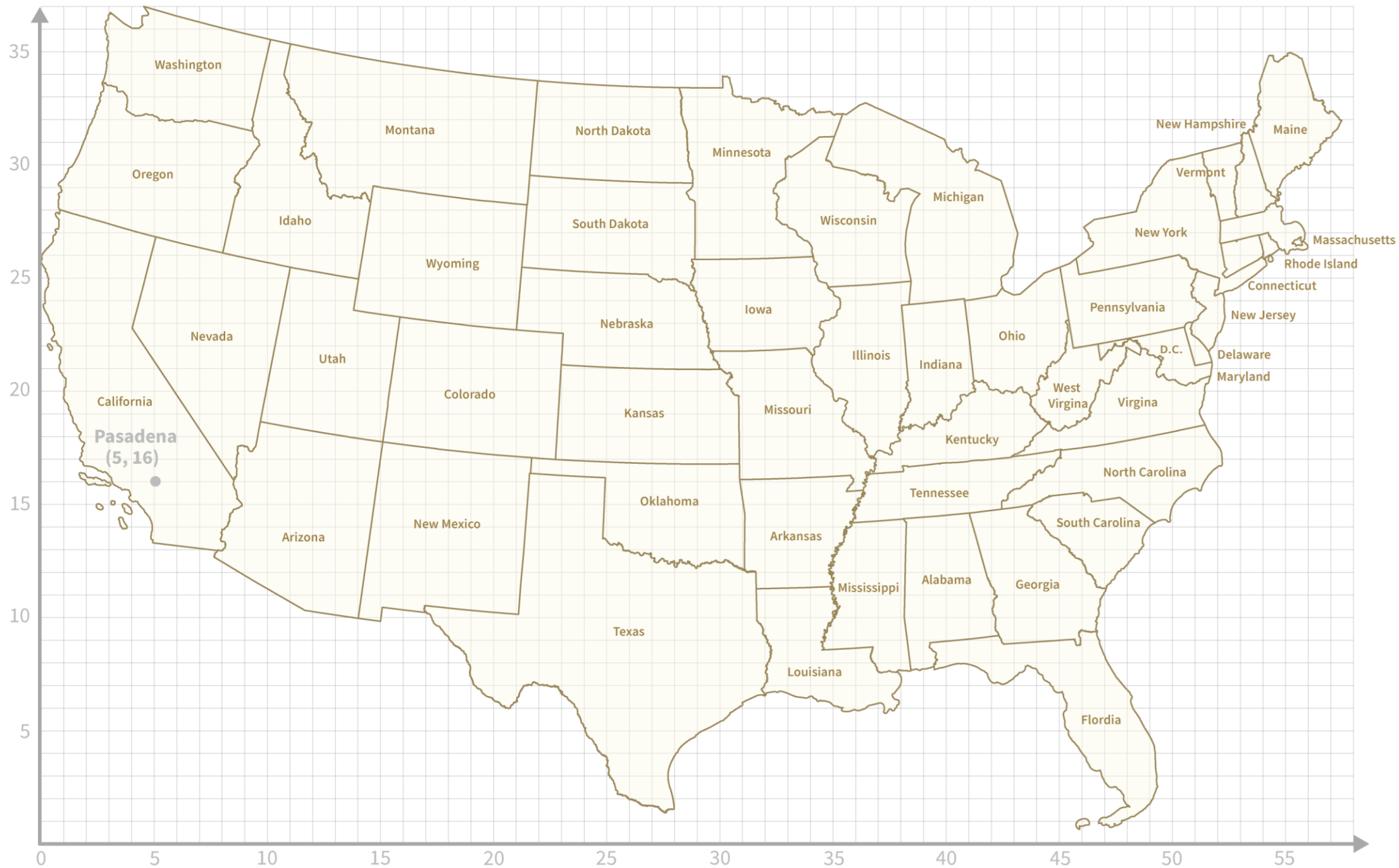
Large mountains immediately behind the city.

Palm trees and other warm-climate vegetation.

Relatively dry-looking conditions.

Several buildings with red tile roofs and Mediterranean/Spanish-influenced architecture.

An older-looking commercial or civic center



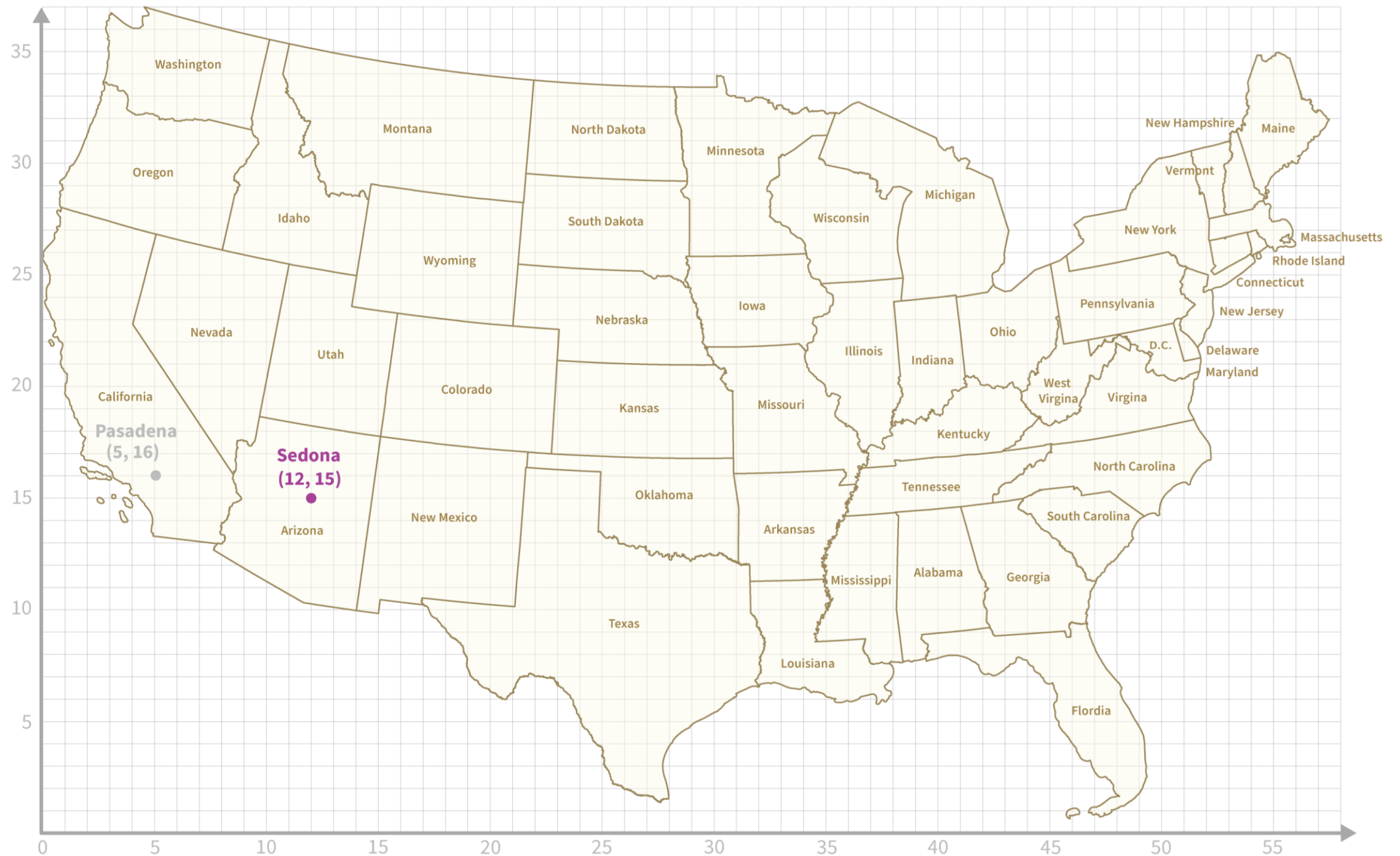
Round 1



Please label it with a purple dot

Sedona, Arizona



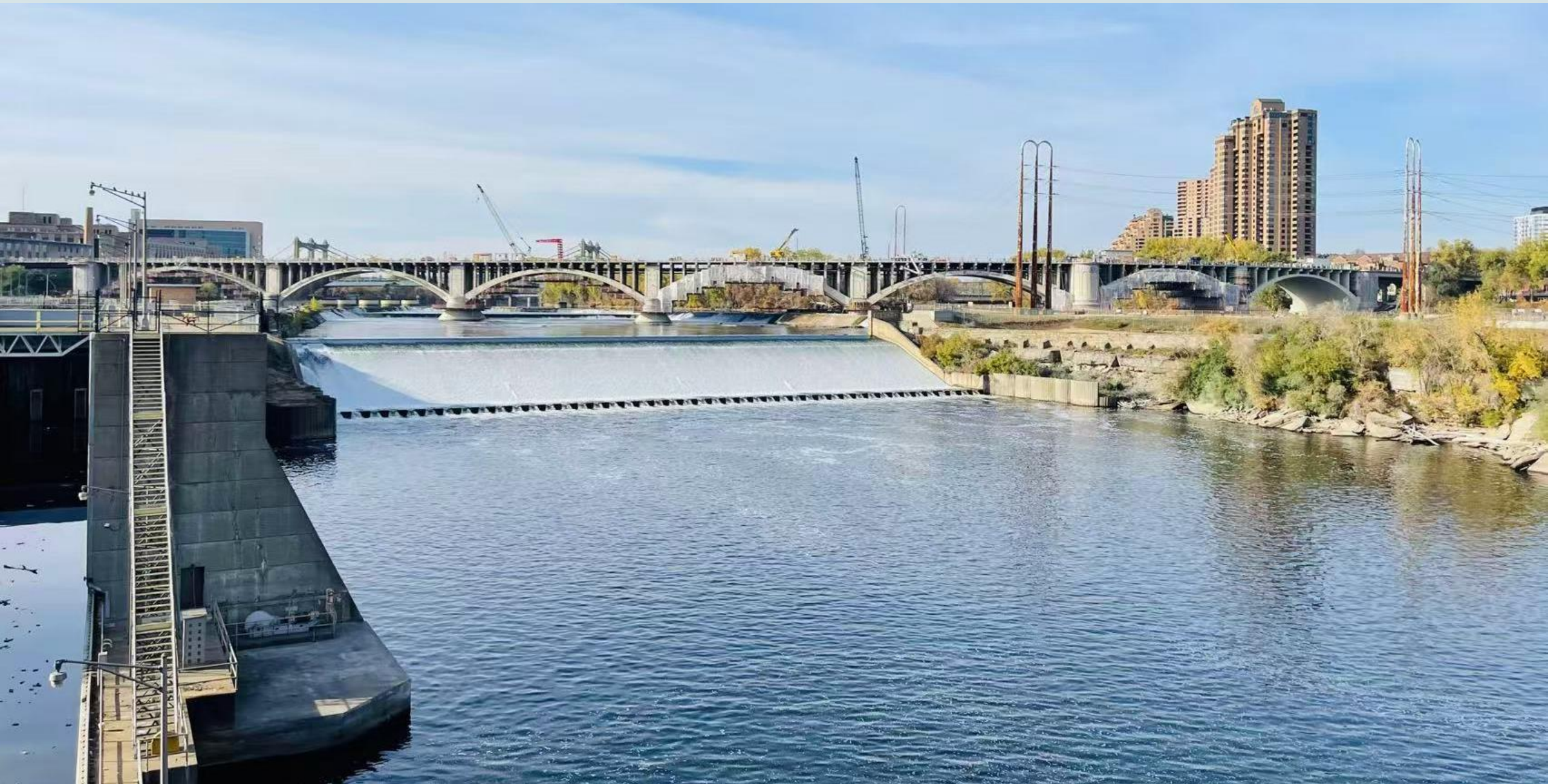


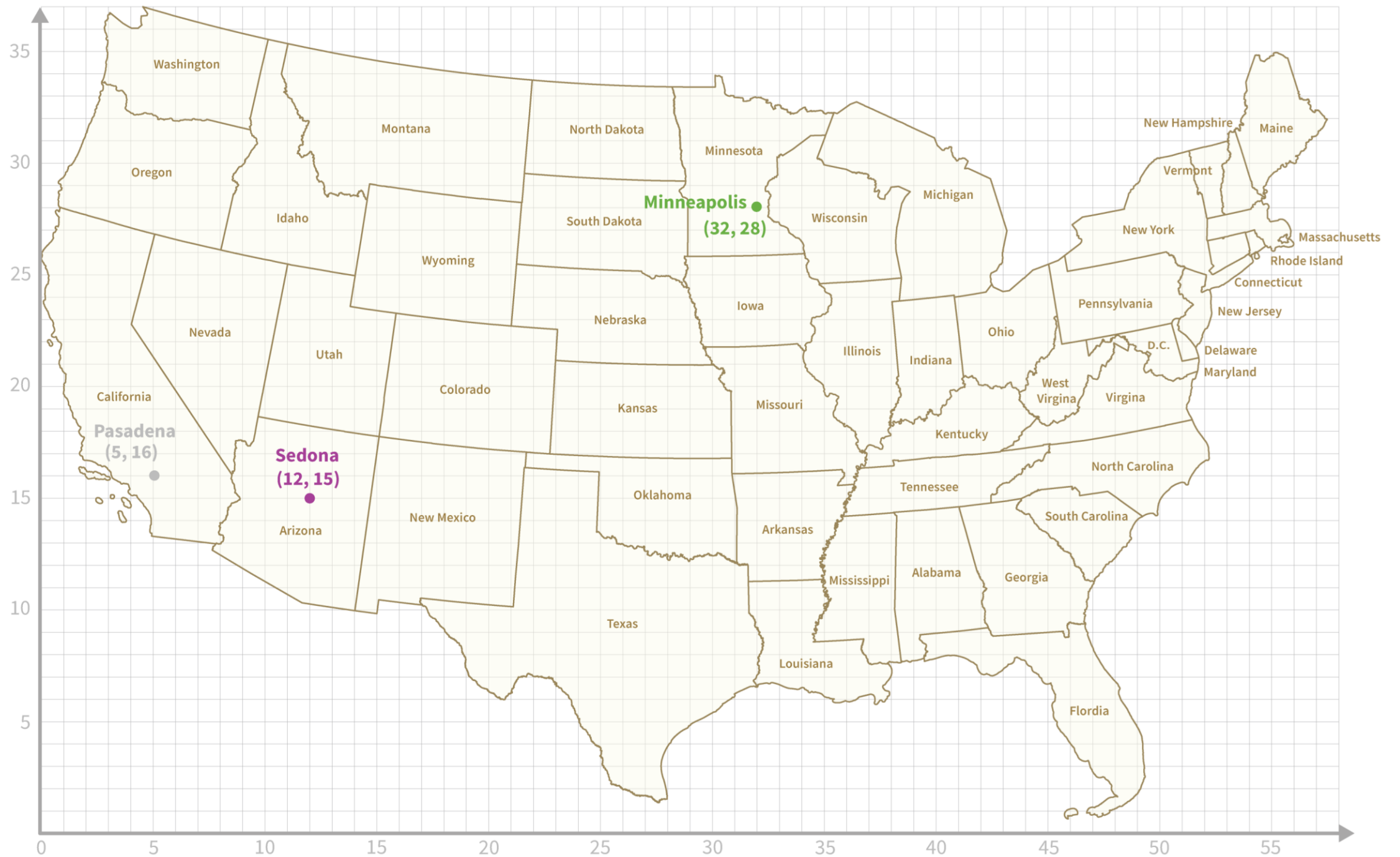
Round 2



Please label it with a green dot

Minneapolis, Minnesota





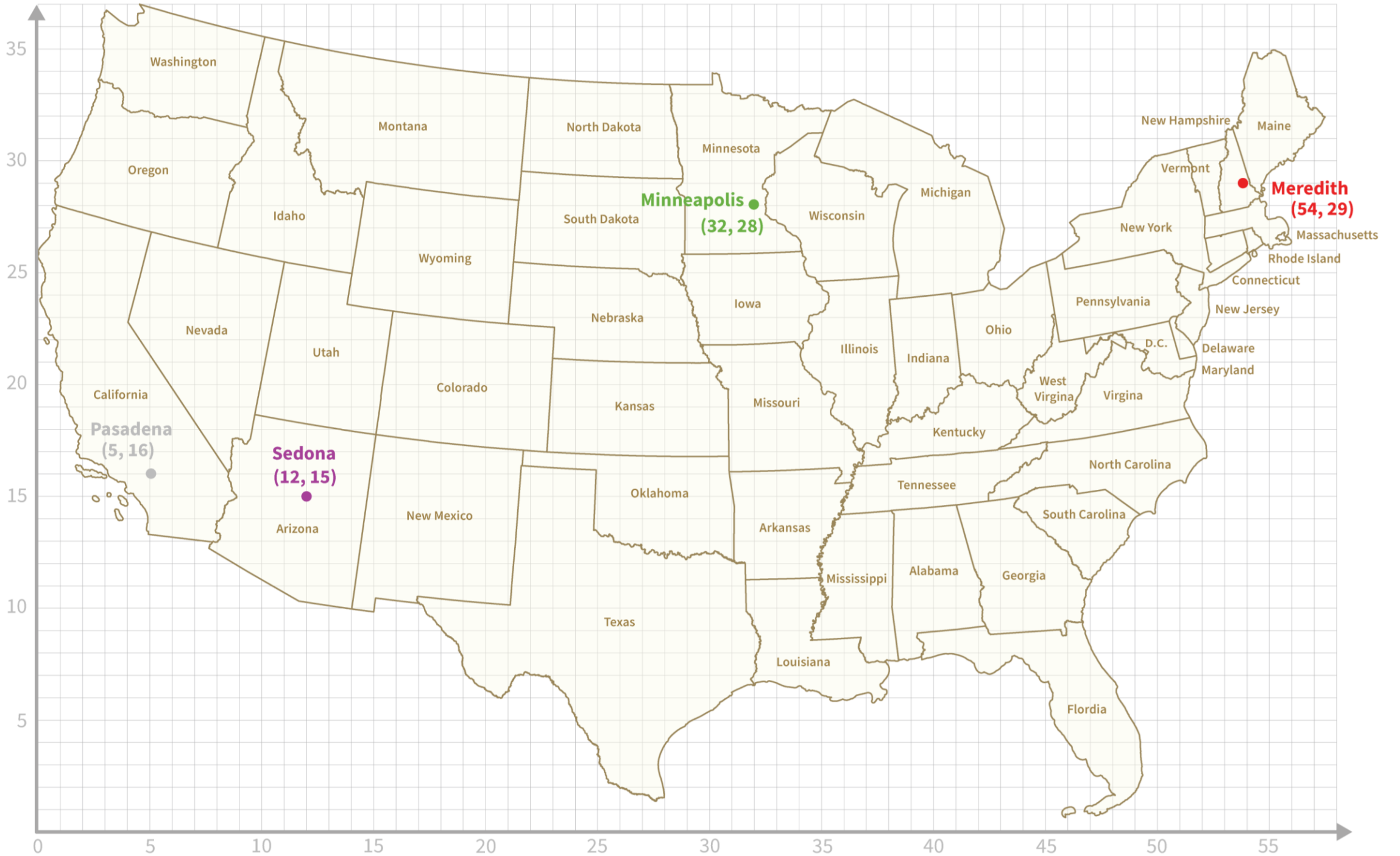
Round 3



Please label it with a red dot

Meredith, New Hampshire





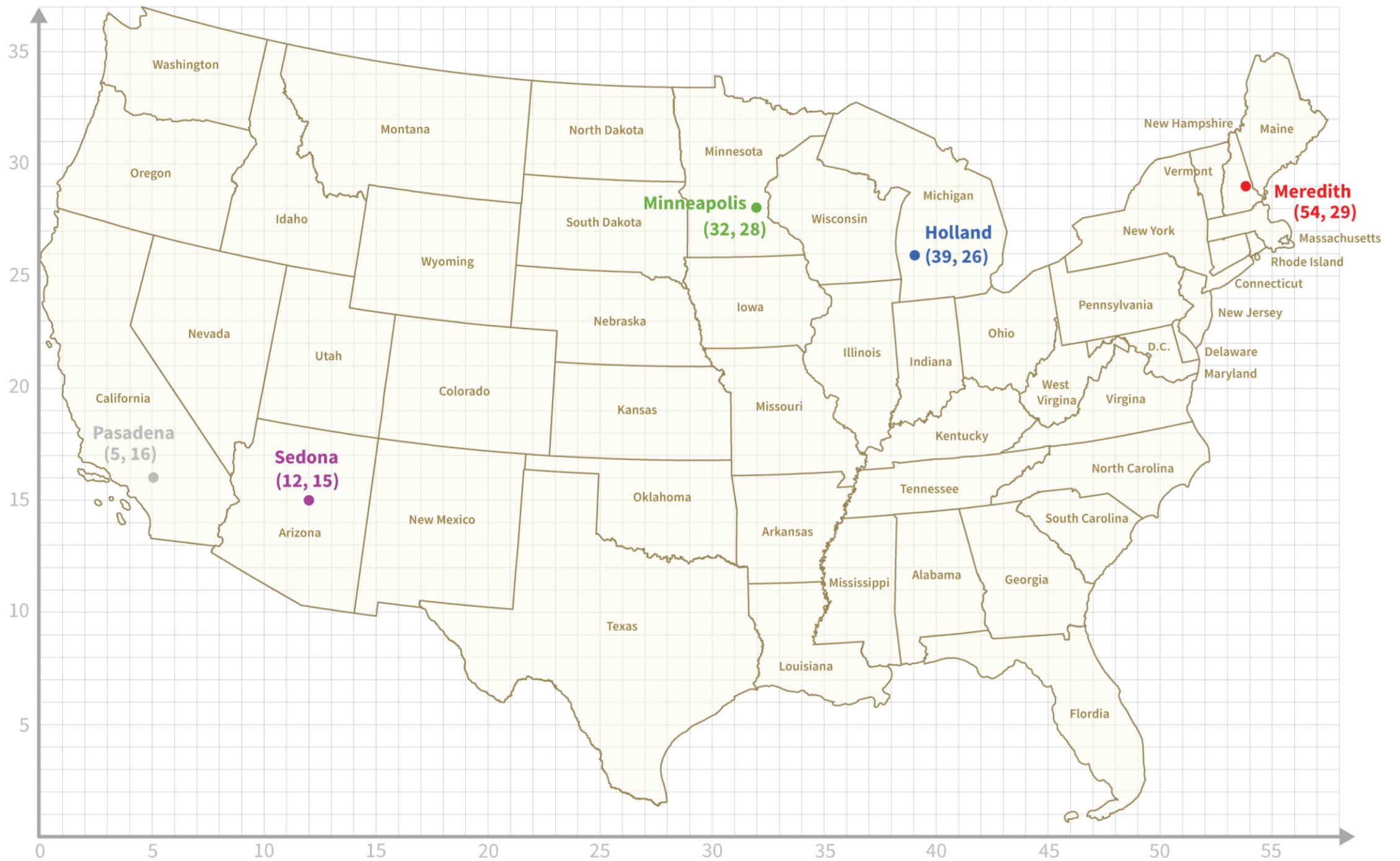
Round 4



Please label it with a blue dot

Holland, Michigan





Debrief

Which image produced the most disagreement, either within your group or across the class?

What types of clues did your group rely on most—landscape, vegetation, architecture, roads, climate, signs, or something else?

Did anyone recognize a place because they had been there, lived nearby, or knew a similar region?

Were there times when two clues seemed to point in different directions? How did your group decide which one to trust?

Were you ever very confident but wrong, or very uncertain but surprisingly close?

**Does anything from today seems consistent with predictions you made last week?
(familiarity with region, travel experience, geographic background)**

WHAT IS SPATIAL DATA

With your guesses, we made spatial data.

Each observation has a location, but that location was produced through a process: the image, the clues available, your prior knowledge, your uncertainty, your group discussion, and the coordinate grid.

Spatial data are observations linked to locations.

To interpret them well, we also need to understand how those observations were generated, i.e. *the data generating process*.

A Quick Note on Patterns

THE IMPORTANCE OF SPATIAL PATTERNS

- Once observations have locations, they create **spatial patterns** (maps from last week, the guesses we made).
- The main approach to geographic analysis is to use those patterns to **reason about the *processes* that produced them.**

$$\langle \mathbf{s}, Z, z(\mathbf{s}) \rangle$$

where

$\mathbf{s}$ = a point in space--time,

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The Cancer Map Mystery

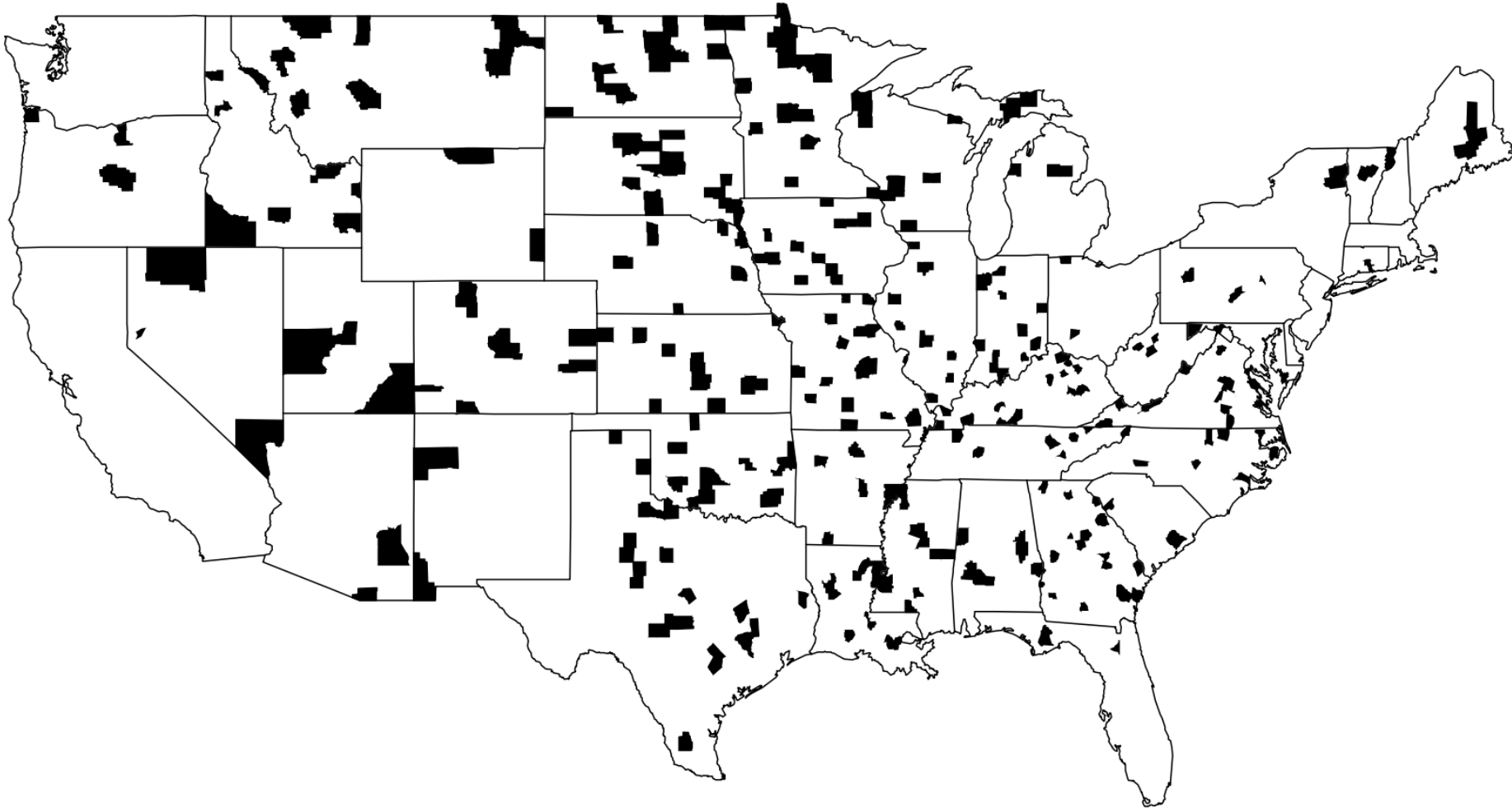
Pattern  **Data Generating Process**



Imagine you are public-health investigators. A suspicious pattern has appeared in U.S. kidney cancer mortality data. Your job is to decide what is going on based on the pieces of information that is provided to you.

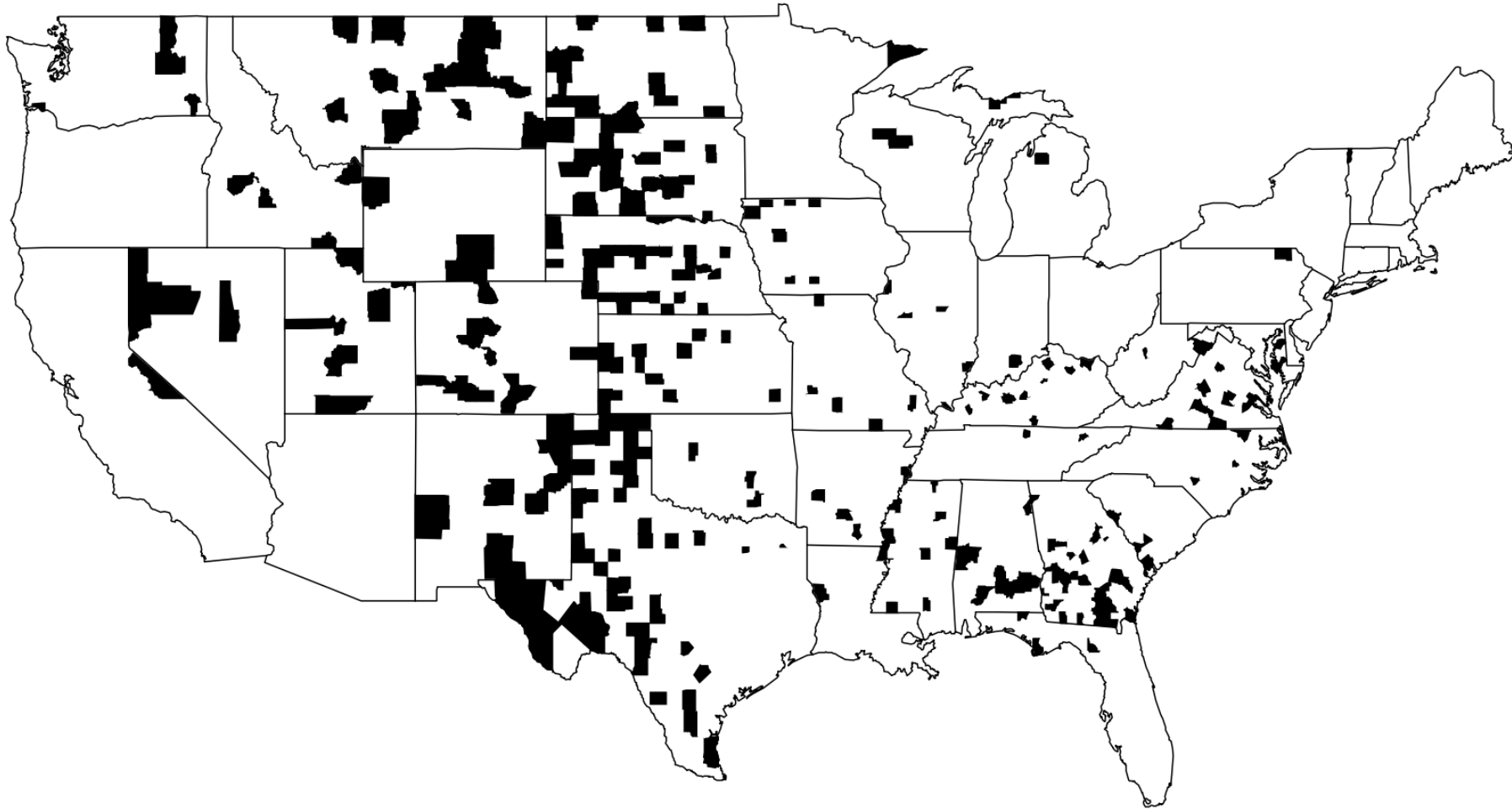
Evidence I

Highest kidney cancer death rates



Evidence 2

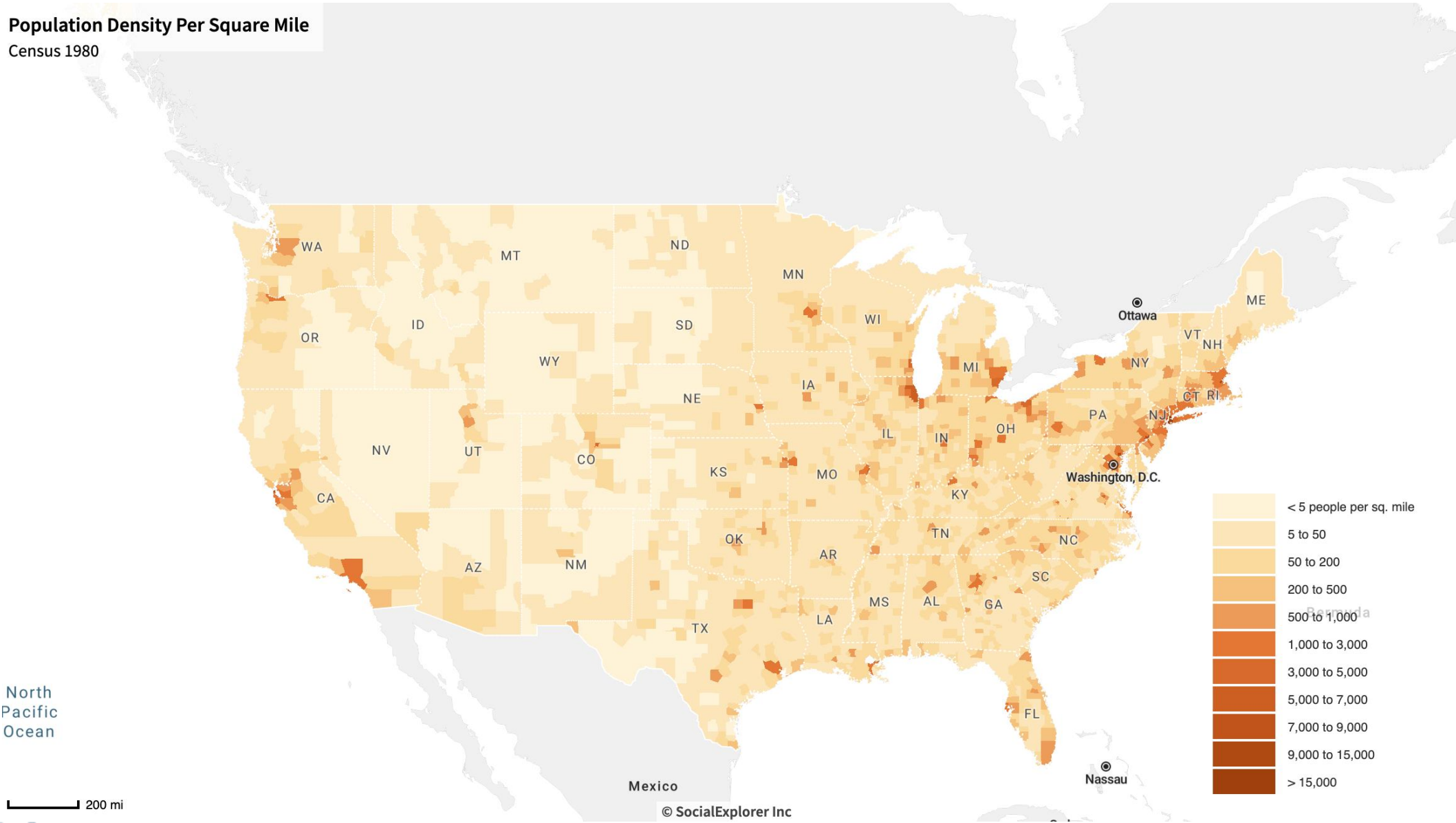
Lowest kidney cancer death rates



Evidence 3

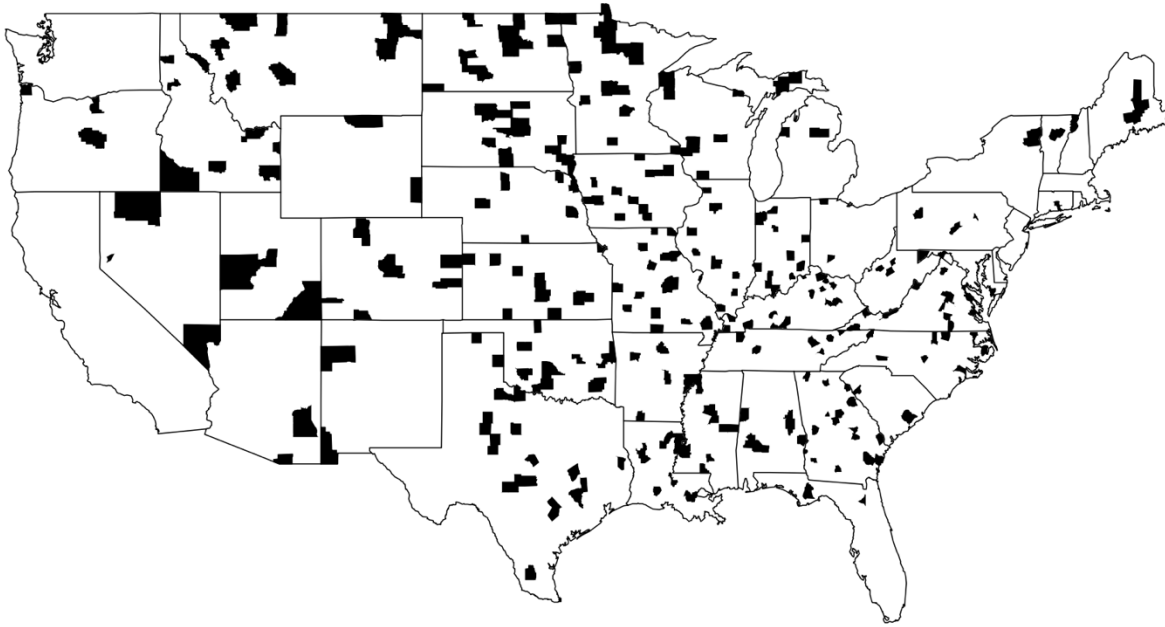
Population Density Per Square Mile

Census 1980

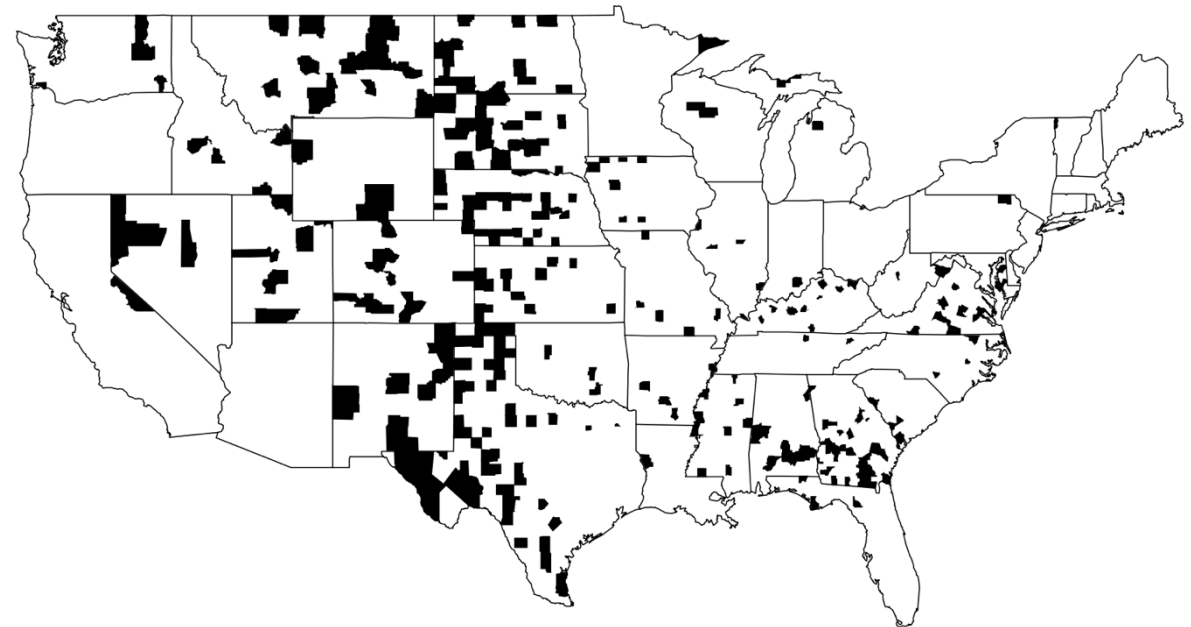


What do many of the extreme-rate counties have in common?

Highest kidney cancer death rates



Lowest kidney cancer death rates



Think about how rates are calculated:

$$\text{Observed Rate} = \frac{\text{Number of Cases}}{\text{Population}}$$

*Maybe it is also showing the effect of small populations and random variation, i.e the **data generating process***

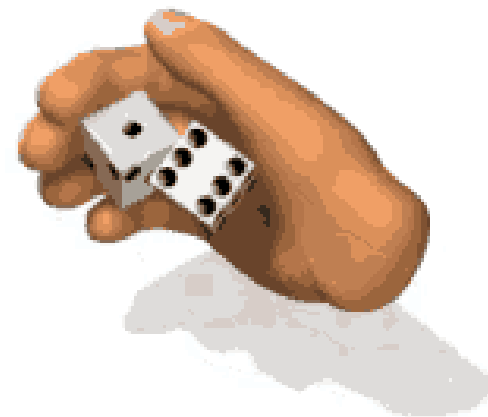
Game Rules

We are going to build a simplified hypothetical cancer map that consists of 14 counties.

In groups of three, each of you will receive a county card containing its hypothetical population size. Each county has different population sizes, but all of them have the same underlying cancer risk

You will roll a die to determine the cancer rate with the following rule:

- Roll a 1 = cancer case
- Roll 2–6 = no case
- Each roll represents one person.



Game Rules

You will roll the dice based on its population size. A county with population 10 rolls 10 times; population 20 rolls 20 times. Whichever group gets Chittenden, Rutland, and Washington county will receive two dices to speed up the process.

You need to count the total number of ones from the dice roll, which represent the number of people getting cancer.

You will repeat this twice

You will report back to the instructor the number of ones for your county (twice). Together as a class, we will calculate the observed rate and visualize them on a map.

At the end of this lecture, each group needs to submit the county card and **write down the name of the people they work with in the group (for attendance :)**



Total Population: 27

Simulation 1

Simulation 2

Debrief

Why do county rates differ if the underlying risk was identical everywhere?

Which counties produced the most extreme rates?

What role might population size be playing?

Is an observed hotspot necessarily evidence of higher underlying risk?

BEHIND THE PATTERN

We consider a county with 1000 people: if it has even one kidney cancer death in the 1980s, its rate is 1 per thousand per year, which is among the highest in the nation.

If a small county has an observed rate of 1 per thousand per year, this is probably random fluctuation, but if a large county such as Los Angeles has a very high rate, it is probably a real phenomenon.

BEHIND THE PATTERN

A map can show real geographic differences, but it can also show random variation, population size, measurement choices, and other features of the data-generating process.

DEMONSTRATION

1000 Simulations of the Same Geography!

Debrief

The histograms show the distribution of possible observed rates that could be generated under the same underlying process.

If we repeat that process many times, we can see which outcomes occur often and which occur rarely. That pattern of possible outcomes and their frequencies is a *probability distribution*.

A Note on Probability Distribution

Each individual die roll in the activity represents a binary outcome, either yes or no. The probability of observing a case is therefore

$$p = \frac{1}{6}$$

For a county with population n , we repeat this same yes/no process n times. The total number of cases observed in that county can therefore be represented using a binomial distribution:

$$X \sim \text{Binomial}(n, p)$$

where n is the county population, p is the underlying probability of a case, and X is the number of cases we happen to observe

A Note on Probability Distribution

For a binomial proportion, the variance of the observed rate is

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

As n increases, this variance decreases. Therefore,

small $n \implies$ greater variability in observed rates

large $n \implies$ more stable observed rates

When $n = 10$, one additional case changes the observed rate by $\frac{1}{10} = 0.10$

When $n = 100$, one additional case changes the observed rate by $\frac{1}{100} = 0.01$

About spatial pattern and process

Geographers often begin with a spatial pattern and ask what process could have produced it. Probability gives us a way to think about the range of patterns that could arise even when the underlying process remains unchanged.

Before explaining a mapped pattern, ask how the data were generated, what the denominator is, how much random variation is possible, and whether the apparent pattern could arise even if the underlying process were the same everywhere.

Takeaways

- Spatial data are observations connected to locations.
- Locations do not appear magically; they are recorded through some process.
- A geographic pattern may reflect an underlying process in the world, but it may also reflect population size, chance, measurement, uncertainty, or the way data were collected.
- GeoGuesser activity showed how people can produce spatial data through observation, inference, prior knowledge, and uncertainty.
- The cancer-map mystery activity showed that dramatic-looking spatial patterns can arise even when the underlying risk is identical everywhere.
- The habit to build this quarter: when you see a spatial pattern, *ask what process could have generated this pattern?*

LOOKING AHEAD

There will be **No Class Thursday**, and you are asked to independently review some statistics you should already be familiar with (e.g., means, popular probability distributions, hypothesis testing). The readings are posted on Canvas.

There will be tutorial this week! The first tutorial is designed to familiarize you with the tutorial structure, R & R Studio, and the computational notebooks we will use during the course. We've prepared this as a platform for you to succeed. Both sections will be led by Die this week. **Nothing is due until Week 3.**

Die will host her office hours starting this week

Emily will host her office hours starting next week.

Lecture slides and demo will be posted on the course website.

If you have a non-confidential/emergent question regarding the lecture, tutorial, etc. please use the GEOG 172 Help Desk Google form to submit questions.